Hill Choy

hill.choy.25@ucl.ac.uk | hillhc | +44 7871 067385, +852 6171 8745 | my website

SUMMARY

AI Software Engineer specializing in end-to-end computer vision pipelines and Edge AI deployments. Proven ability to translate complex machine learning research into robust, operational backend systems that automate physical workflows and drive sustainable optimization for commercial and public sector clients.

EDUCATION

University College London (UCL)
MSc Computer Graphics, Vision and Imaging (***Current Average: 79%***) — *Sept 2025 - Present*
- **Highlighted Coursework**: Machine Vision, ML for Visual Computing, Inverse Problems, Robot Vision.

Hong Kong University of Science and Technology (HKUST)
BEng in Computer Engineering (***Second Class Honors, Division 1***) — *Sept 2020 - Jul 2024*
- **Highlighted Coursework**: Intro to NLP, Computer & Communication Networks.

IELTS Academic
8.0 (*Reading: 8.5, Listening: 9.0, Writing: 7.0, Speaking: 7.0*)

RELEVANT EXPERIENCE

Hyperspectral Image Analysis for Safe-Fabric Segmentation — **United Kingdom**
UCL, MSc Industry Dissertation Project (*Partner:* ***Zori Tex***) — *June 2026 – Sept 2026* (*Expected*)
- Collaborating with industry partner Zori Tex to research and develop an automated computer vision pipeline for textile waste sorting utilizing hyperspectral imaging.
- Aiming to explore PCA and shallow networks to condense high-dimensional data for memory-efficient CNN processing, alongside a weakly supervised Autoencoder approach to handle uneven real-world illumination.

Research Assistant — **Hong Kong**
Prof. Shueng-Han Gary Chan, CSE, HKUST — *Aug 2024 - Jun 2025*
- **K11 MUSEA Cars and Pedestrian Flow Monitoring System**
  - Engineered an end-to-end analytics system tracking 60,000+ daily visitors, utilizing OpenCV and a PyTorch-based YOLO model for real-time local flow counting on edge platforms.
  - Developed a multithreaded Flask backend utilizing Redis for real-time data synchronization and SQLite/Pandas for robust storage, powering a web dashboard to optimize mall operations.
- **Privacy-Preserving Elderly Monitoring with a Novel Edge AI Camera**
  - Contributed to the development of a privacy-preserving Edge AI fall-detection system deployed on an NPU-accelerated OrangePi, utilizing Human Pose Estimation for real-time patient monitoring.
  - Developed rule-based post-processing algorithms to minimize false alarms without compromising true positive cases, achieving 98% accuracy and securing the Silver Award at the Hong Kong ICT Awards 2024.
- **Smart Traffic Management System (Kung Tong, Hong Kong)**
  - Trained and deployed YOLO models on a dual-RTX 4090 server to process live streams from 33 cameras across 11 major junctions, monitoring 20,000+ daily vehicles per junction.
  - Engineered automated client scripts using Python requests to securely transmit vehicle counts and yellow-box stop detections to a centralized API, handling Keycloak OAuth2 token authentication.

Advanced Video Analytics and Edge AI for Smart Carpark Systems — **Hong Kong**
HKUST, Bachelor Degree Final Year Project (*Supervisor:* ***Prof. Shueng-Han Gary Chan, CSE, HKUST***) — *Sept 2023 - Apr 2024*
- Designed an Edge AI smart car park system to track real-time space availability, utilizing OpenCV for video frame grabbing and preprocessing to handle complex feeds from custom low-light and wide-angle cameras.
- Deployed a PyTorch-based YOLOv8 model on a Raspberry Pi by integrating a MobileNetV3 backbone, converting to ONNX format, and utilizing Ultralytics for custom dataset annotation, halving inference latency and reducing parameters by 30% while maintaining baseline accuracy.

Intern (Summer full-time, then part-time) — **Hong Kong**
Hong Kong Telecom (HKT), Integrated Service Provision Center — *June 2023 - April 2025*
- Engineered a Python GUI executable to replace a legacy system and eliminate manual checks, automating both Excel and physical report generation for phone line data.
- Bypassed the lack of a native API using a custom web-scraping pipeline, dynamically extracting interface records to synchronize data with a MySQL database.